505_L02 Numerical Solution to Riccati Equation

I Euler's Method

$\dot{z} = g(x)$ then $z(t+\delta t) \approx z(t) + \delta t\, g(z(t))$

$z_0 = z(t_0)$ ← Initial Value Problem

↳ We know $P(t_1) = S$ ← Final value

A) Final Value Problem

$z_1 = z(t_1)$ $\quad z(t-\delta t) \approx z(t) - \delta t\, g(z(t))$

B) Algorithm For Finding $P(t)$

1) Initialize:
2) $t = t_1$
3) $P = S$
4) while $t \geq 0$
5) $P(t-\delta t) = P(t) - \delta t \overbrace{\left(-A^T P - PA - Q + PBR^{-1}B^T P\right)}^{\dot{P}}$
6) $t = t - \delta t$
7) end

However, direct integration of RE is known to be unstable.

II Use Linear System Theory
Consider the following LTV System

$$\dot{z} = \underbrace{(A - BR^{-1}B^T P(t))}_{M(t)} z = M(t) z$$

$$\Rightarrow \quad z(t) = \Phi(t, t_0)\, z(t_0)$$

$$\Rightarrow \quad \frac{d}{dt}\left(\Phi(t, t_0)\right) = M(t)\, \Phi(t, t_0)$$

A) Matrix Solution

$$\dot{\underline{X}} = \left(A(t) - B(t) R^{-1} B^T(t) P(t)\right) \underline{X}(t)$$

$$\underline{X}(T) = I, \quad \underline{X} \in \mathbb{R}^{n \times n}$$

$$\underline{Y}(t) = P(t)\, \underline{X}(t) \quad \text{or} \quad P(t) = \underline{Y}(t)\, \underline{X}^{-1}(t)$$

where $\underline{X}^{-1}$ exists because $\Phi(t, t_0)$ is full rank

B) Time Derivative

$$\dot{\underline{Y}} = \dot{P}\underline{X} + P\dot{\underline{X}}$$

$$\Rightarrow \quad \dot{\underline{Y}} = \underbrace{\left(-A^T P - PA - Q + PBR^{-1}B^T P\right)}_{\dot{P}} \underline{X} + P \underbrace{\left(A - BR^{-1}B^T P\right) \underline{X}}_{\dot{\underline{X}}}$$

After expanding and simplifying we get

$$\dot{\underline{Y}} = -A^T \underbrace{P\underline{X}}_{\underline{Y}} - Q\underline{X} = -A^T \underline{Y} - Q\underline{X}$$

C) Replace RE with linear system:

$$\dot{\bar{X}} = A\bar{X} - BRB^T\bar{Y}$$

$$\dot{\bar{Y}} = -Q\bar{X} - A^T\bar{Y}$$

$$P = \bar{Y}\bar{X}^{-1} \qquad P \in \mathbb{R}^{n\times n}$$

$$\bar{X}(T) = I \qquad \bar{X} \in \mathbb{R}^{n\times n}$$

$$\bar{Y}(T) = S \qquad \bar{Y} \in \mathbb{R}^{n\times n}$$

Integrating this linear system is much more stable

D) Solve RE with new Linear System

$$\begin{bmatrix} \dot{\bar{X}} \\ \dot{\bar{Y}} \end{bmatrix} = \underbrace{\begin{bmatrix} A & -BRB^T \\ -Q & -A^T \end{bmatrix}}_{M} \begin{bmatrix} \bar{X} \\ \bar{Y} \end{bmatrix}$$

If the system is LTI there is no time dependence.

1) LTI

$$\begin{bmatrix} \bar{X}(t) \\ \bar{Y}(t) \end{bmatrix} = e^{M(t-T)} \begin{bmatrix} I \\ S \end{bmatrix}$$

$$P(t) = \bar{Y}(t)\bar{X}^{-1}(t)$$

Very small Numerical Error

III Example

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Drive x_1 to origin using as little energy as possible

A) Formulate Cost

$$\min_{u(\cdot)} = J(u(\cdot)) = \int_0^T u^2(t)\,dt + x_1^2(T)$$

$$Q = 0 \quad R = 1 \quad S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$BRB^T = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

B) Integrate Linear System

$$\bar{X} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

stated Previously

$\uparrow$

$\bar{X}(T) = I$

$$\dot{\bar{X}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

$x_{11}(T) = 1,\ x_{22}(T) = 1 \quad x_{12}(T) = x_{21}(T) = 0$

$$\bar{Y}(T) = -Q\bar{X} - \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

$$\bar{Y}(T) = S$$

$$y_{11}(T) = 1, \quad y_{12}(T) = y_{21}(T) = y_{22}(T) = 0$$

c) For this example it is possible to solve

$y_{11} = 1 \quad y_{12} = 0 \qquad x_{11} = 1 + \frac{1}{6}(t-T)^3$

$y_{21} = T-t \quad y_{22} = 0 \qquad x_{21} = \frac{1}{2}(t-T)^2$

$x_{12} = t - T$

$x_{22} = 1$

$$\Rightarrow P(t) = \bar{Y}(t)\bar{X}^{-1}(t) = \frac{1}{1+\frac{1}{3}(T-t)^3} \begin{bmatrix} 1 & T-t \\ T-t & (T-t)^2 \end{bmatrix}$$

$$\Rightarrow u(t) = -R^{-1}B^T P(t)x(t) = \underbrace{-1\begin{bmatrix} 0 & 1 \end{bmatrix} P(t)x(t)}_{\text{Optimal Control}}$$